Future-Horizon Regularity and One-Sided Radiative Traces Do Not Select the Einstein Sector of Pure Weyl Gravity

GPT-5.6.sol* Claude Fable 5†

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Abstract

Maldacena-type boundary selection isolates an Einstein sector of conformal gravity in Euclidean anti-de Sitter or late-time de Sitter settings. We ask whether future-horizon regularity and one-sided Lorentzian radiative trace conditions do the same for linear perturbations of a Schwarzschild black hole in four-dimensional pure Weyl gravity. They do not.

Modulo the four-dimensional Euler boundary term, the Hessian of the Weyl-squared action about any Ricci-flat background factors through the trace-adjusted linearized Ricci tensor q_{ab}, which obeys a constrained linearized Einstein equation. Its target-gauge sector is a Maxwell field with the opposite bulk kinetic sign to ordinary Maxwell theory. The Einstein kernel is consequently a universal null subspace of the bulk Hessian. Its literal C^2 self-current is an Euler cut current, so the complete smooth compact-frequency Einstein wave-packet space is totally isotropic. Any nondegenerate restriction of the inherited form that retains an Einstein direction is nevertheless indefinite because populated additional modes pair nontrivially with it.

In the axial $\ell = 2$ sector we identify an exact filtered differential module with successive quotients

$$M^{(2)}_{\mathrm{RW}}, \qquad M^{(2)}_{\mathrm{RW}}, \qquad M^{(1)}_{\mathrm{RW}}.$$

The two spin-two factors form a non-split self-extension. The action-derived Lee–Wald form on the incoming trace space is nondegenerate with signature $(1, 2)$. In factor coordinates it is congruent, for every real $\omega > 0$, to

$$\pi\alpha_{\mathrm{W}} \begin{pmatrix} 0 & 576\omega/5 & 0 \\ 576\omega/5 & 0 & 0 \\ 0 & 0 & -32/(15\omega) \end{pmatrix}.$$

Scalar Regge–Wheeler Wronskians and boundary dévissage prove that the map from future-horizon-regular data to this complete incoming Krein space is invertible for every positive real frequency. Negative-flux directions are therefore populated by regular exterior solutions, rather than being formal endpoint artefacts.

For every integer $\ell \geq 2$ and every real $\omega > 0$, the scalar spin-two Regge–Wheeler and spin-one Maxwell differential modules are simple over $\mathbb{C}(r)[D]$, with rational endomorphism ring $\mathbb{C}$. Together with specialization-safe nonsplitting of the axial $\ell = 2$ self-extension, this implies that its complete local commutant is the dual-number algebra and its only rational local dynamical involutions are $\pm I$; neither can make the hyperbolic spin-two form positive. A

*OpenAI model and principal programme author.

†Anthropic model and computational coauthor. The project was commissioned and coordinated by Asger Alstrup Palm (`asger@area9.dk`), who initiated the questions, exercised editorial control, and serves as corresponding human contact, but claims no technical authorship.

nonlocal spectral fundamental symmetry does exist on each positive-real-frequency fiber and on every compact band, but its natural all-frequency completion is threshold-weighted.

On the certified cell $\omega \in [0.49995, 0.50005]$ the outgoing and future-horizon spaces are matching three-channel Krein spaces, the outgoing map is invertible, and the one-sided scattering map obeys the exact integrated pseudo-isometry. Analytic continuation gives outgoing population on an open dense, full-measure subset of the positive axis; possible failures are isolated scalar reflection zeros. The same filtration excludes exponentially growing separated axial modes.

Finally, a validated projective Evans calculation encloses one simple damped spin-two quasinormal frequency and certifies a nonzero intrinsic extension selector. The spin-one factor is a local unit there. The complete connection therefore has Smith valuations $(0, 0, 2)$, algebraic multiplicity two, geometric multiplicity one, and one length-two connection root chain. This is a connection-level intrinsic exceptional point. A second-order physical Green-resolvent pole remains conditional on a separate analytic Fredholm realization.

1 Question, answer, and scope

Pure Weyl gravity is defined by

$$S_{\mathrm{W}}[g] = \alpha_{\mathrm{W}} \int_M \sqrt{-g}\, C_{abcd} C^{abcd}\, \mathrm{d}^4 x, \qquad B_{ab} = 0. \tag{1}$$

Every Einstein metric is Bach-flat, but the fourth-order Bach equation has additional local solutions. The question addressed here is deliberately narrow:

> Do future-horizon regularity and standard one-ended incoming or outgoing real-frequency trace conditions at null infinity force a Schwarzschild perturbation of pure Weyl gravity into the Einstein conformal sector?

For the complete axial $\ell = 2$ system the answer is no. The extra curvature carrier survives these one-ended conditions, participates in the action-derived current, and supplies populated negative directions in a nondegenerate Krein space. The title theorem is a scattering statement: at real frequency, future-horizon-regular solutions generally have both incoming and outgoing null-infinity traces. It does *not* assert a non-Einstein separated solution that is simultaneously future-horizon regular and purely outgoing at infinity. That two-ended condition is the quasinormal boundary problem and is kept logically separate below.

Definition 1.1 (Radiative endpoint class tested here)**.** The tested class consists of phase-factored solutions that are analytic future-horizon ingoing germs, extend through the Schwarzschild exterior, and have finite factor-normalized Regge–Wheeler/Maxwell traces at null infinity. On the real axis we study either one-sided trace map T_- or T_+. "Outgoing population" means surjectivity onto the outgoing trace; it does not mean that the incoming trace vanishes. The complex-frequency theorem uses its separately declared future-horizon-regular/pure-outgoing quasinormal class. We do not rule out stronger curvature conditions, mixed field/normal-derivative conditions, or the direct carrier constraint $q_{ab} = 0$.

This is not a statement that no boundary condition can select the Einstein sector. The local Cauchy restriction

$$\delta R_{ab}|_{\Sigma} = 0, \qquad \nabla_n \delta R_{ab}|_{\Sigma} = 0$$

does select the linearized Einstein kernel in the axial carrier problem. Moreover, the vacuum Einstein metrics form an exact nonlinear Bach-flat submanifold, so the Einstein equations define an algebraically consistent nonlinear truncation. What remains open is narrower: whether the Ricci-zero Cauchy restriction determines a unique gauge-fixed Bach development and whether a differentiable physical phase space selects that restriction rather than imposing it externally. Nor is this paper a quantum-positivity theorem. "Negative" below always means negative with respect to the classical Lee–Wald flux form.

1.1 Five principal results

The paper is organized around five statements.

1. On every four-dimensional Ricci-flat background, the bulk pure-Weyl Hessian modulo the Euler term factors through linearized Ricci curvature. The trace-adjusted carrier obeys a constrained linearized Einstein equation; its target-gauge sector is wrong-sign Maxwell. The Einstein wave-packet space is totally isotropic but nonradical, and no nondegenerate semidefinite restriction can retain its line.

2. The complete axial system has an exact RW/RW/spin-one filtration, with a nontrivial repeated-spin-two extension and no rational local spin-one/spin-two intertwiner. The repeated block has dual-number local commutant and no nontrivial semisimple local branch observable. Its associated graded pieces are the source Einstein graviton, the physical target Einstein carrier, and the Maxwell target-gauge vector.

3. Its incoming action-flux space has signature $(1, 2)$, and every incoming trace is realized at every $\omega > 0$ by a future-horizon-regular solution. A spectral fundamental symmetry gives a positive compact-band norm, with exact ω, ω, ω^3 factor weights at threshold.

4. On a certified nonzero frequency cell every outgoing trace is likewise realized by a future-horizon-regular solution; this is a one-ended surjectivity statement, not a pure-outgoing quasinormal statement. Outgoing surjectivity is generic on the positive real axis, and the filtered system has no exponentially growing separated axial mode.

5. One enclosed damped spin-two quasinormal frequency is a certified connection-level exceptional point with Smith valuations $(0, 0, 2)$.

1.2 Controlled vocabulary

Definition 1.2. In a fixed conformal gauge, the *Einstein kernel* is the kernel of the linearized Ricci map $h \mapsto \delta R_{ab}[h]$. Gauge-invariantly, the *Einstein conformal sector* is the kernel of the carrier class

$$\mathfrak{q}[h] = \left[\delta R_{ab}[h] - \frac{1}{6} g_{ab} \delta R[h]\right],$$

where the brackets quotient target Einstein gauge transformations induced by source Weyl transformations. Source diffeomorphisms leave this carrier zero on a Ricci-flat background, while $h \mapsto h + 2\sigma g$ sends $q_{ab} \mapsto q_{ab} - 2\nabla_a \nabla_b \sigma$. Thus $\mathfrak{q}[h]$ is invariant under the allowed source diffeomorphism/Weyl gauge. Its target-gauge part is represented by $F_{ab} = 2\nabla_{[a}\eta_{b]}$; $F \neq 0$ is a direct gauge-invariant witness that $\mathfrak{q}[h] \neq 0$. *Population* of an endpoint trace space means surjectivity of the relevant horizon-to-endpoint connection map. *Factor frame* means a basis adapted to the differential-module filtration. A *signature frame* diagonalizes a Hermitian flux form. These two frames are not identified.

The result tags used below have the repository meanings LOCAL-ALGEBRAIC, REDUCED-MODE, and LORENTZIAN-CAUSAL. A reduced-mode theorem is never used as evidence for a complete Lorentzian quantum or time-domain construction.

2 Relation to existing work

Maldacena's conformal-gravity construction uses a Neumann condition in Euclidean AdS or at late times in de Sitter space to select Einstein solutions [1]. The present problem has a different signature, background, and boundary geometry: a one-ended Lorentzian Schwarzschild exterior with a future event horizon and null infinity.

The abstract operator complex is established in the conformal-geometry literature. Branson and Gover constructed the conformal deformation detour complex on obstruction-flat, and hence Bach-flat in four dimensions, backgrounds [2]. Gover, Somberg, and Souček embedded such constructions in Yang–Mills detour complexes [3]. Deser, Joung, and Waldron wrote the nonlinear Bach factorization through Einstein and Schouten operators and discussed the corresponding local tensor/vector content [4]. We therefore do not claim the abstract Schouten–Bach factorization, detour complex, or local two-tensor-plus-vector count as new.

Ricci-tensor variables have long been useful for exposing higher-derivative spin-two sectors. Recent quadratic-gravity work computes axial quasinormal modes of massless and massive spin-two fields [10], develops a Ricci-tensor treatment of Kerr perturbations [11], evolves Einstein–Weyl perturbations in time [12], and studies the excitation and suppression of beyond-GR ringdown by point-particle sources [13]. A recent thesis gives a complementary radial Ricci-tensor reconstruction and its relation to the Schwarzschild–Bach family [14].

The distinctive result here is the explicit Schwarzschild realization of the detour structure as a non-split RW/RW/Maxwell filtration, together with its action-derived Krein form, exact horizon and incoming population, and the failure of ordinary Lorentzian endpoint conditions to select the Einstein kernel. The recent black-hole works above concern quadratic or Einstein–Weyl theories, massive sectors, radial reconstructions, or observable excitation. The claims are complementary: we do not infer excitation amplitudes or a complete waveform from endpoint population alone.

3 Universal Ricci factorization and the Bach complex

In four dimensions,

$$C_{abcd}C^{abcd} = E_4 + 2R_{ab}R^{ab} - \frac{2}{3}R^2, \qquad E_4 = R_{abcd}R^{abcd} - 4R_{ab}R^{ab} + R^2. \tag{2}$$

The integral of E_4 has only boundary and corner variation. Around a Ricci-flat background define

$$\mathcal{R} : h_{ab} \longmapsto \delta R_{ab}[h], \qquad \Pi(\psi)_{ab} = \psi_{ab} - \frac{1}{3}g_{ab}\psi.$$

Theorem 3.1 (Ricci-factorized pure-Weyl Hessian)**.** *Let g be any four-dimensional Ricci-flat metric. Modulo the Euler boundary/corner term, the mixed Hessian of the pure-Weyl action is*

$$\delta_1\delta_2 S_{\mathrm{W}} = 4\alpha_{\mathrm{W}} \int_M \sqrt{-g}\, \left(\psi_{1,ab}\psi_2^{ab} - \frac{1}{3}\psi_1\psi_2\right) \mathrm{d}^4x. \tag{3}$$

Equivalently, up to the overall sign fixed by the Euler–Lagrange convention,

$$\delta B \sim \mathcal{R}^\dagger \Pi \mathcal{R}.$$

In particular, the linearized Einstein kernel lies in the radical of the bulk Hessian:

$$\mathcal{R}h_E = 0 \quad \Longrightarrow \quad \delta^2 S_{\mathrm{W,bulk}}(h_E, h) = 0 \quad \textit{for every } h.$$

Proof. Substitute (2) into the action. On a Ricci-flat background all terms involving a background Ricci tensor, scalar curvature, or the first variation of the volume density vanish. Polarizing the remaining quadratic term gives (3). Formal integration by parts gives the normal-operator form. □

This explains why Einstein isotropy is neither an axial nor a Schwarzschild accident. The result holds around every four-dimensional Ricci-flat background. It is deliberately a bulk statement modulo the Euler density: the literal C^2 Lee–Wald representative can retain an exact Euler or corner contribution. That caveat is compatible with the unequal-frequency Einstein current being a nonzero exact expression while the Hermitian Einstein self-flux vanishes.

Theorem 3.2 (Schouten–Einstein carrier factorization)**.** *On every four-dimensional Ricci-flat background, define*

$$q_{ab}[h] = \delta R_{ab}[h] - \frac{1}{6} g_{ab}\delta R[h].$$

Then

$$\delta B_{ab}[h] = -\delta G_{ab}[q[h]], \qquad q = \frac{1}{3}\delta R[h], \qquad \nabla^a q_{ab} = \nabla_b q. \tag{4}$$

Moreover $q[h] = 0$ if and only if $\delta R_{ab}[h] = 0$. Hence the linearized solution spaces fit into the covariant exact sequence

$$0 \longrightarrow \mathcal{H}_E \longrightarrow \mathcal{H}_B \xrightarrow{\mathcal{Q}} \{q : \delta G[q] = 0,\ \nabla^a q_{ab} = \nabla_b q\} \cap \operatorname{im}\mathcal{Q} \longrightarrow 0.$$

Proof. The linearized Bianchi identity gives $\nabla^a \delta R_{ab} = \nabla_b \delta R/2$, from which the trace and divergence constraints follow. Substitution into the general Ricci-flat formula for $\delta G[q]$ reproduces (8) coefficient by coefficient. Taking the trace of $q = 0$ gives $\delta R = 0$, and hence $\delta R_{ab} = 0$. □

Theorem 3.3 (Second-order parent system and factorized current)**.** *On a four-dimensional Ricci-flat background, the bulk quadratic pure-Weyl action is equivalent, modulo the Euler boundary/corner term, to*

$$S^{(2)}_{\text{par}}[h,f] = 4\alpha_{\text{W}} \int_M \sqrt{-g}\left[f^{ab}\delta G_{ab}[h] - \frac{1}{2}\left(f_{ab}f^{ab} - f^2\right)\right] \mathrm{d}^4x, \tag{5}$$

where f_{ab} is an independent symmetric tensor. Its Euler–Lagrange equations are

$$\delta G_{ab}[f] = 0, \qquad f_{ab} = q_{ab}[h]. \tag{6}$$

Let $\boldsymbol{\omega}_E(a,b)$ denote the antisymmetric Green current of the linearized Einstein operator. Then

$$\boldsymbol{\omega}_{\text{W}}\big((h_1,f_1),(h_2,f_2)\big) = 4\alpha_{\text{W}}\left[\boldsymbol{\omega}_E(h_1,f_2) + \boldsymbol{\omega}_E(f_1,h_2)\right] + dk_{E_4}. \tag{7}$$

Proof. Variation of (5) with respect to f gives

$$\delta G_{ab}[h] = f_{ab} - g_{ab}f.$$

Since $\delta G_{ab}[h] = q_{ab} - g_{ab}q$, the trace equation in four dimensions gives $f = q$, and hence $f_{ab} = q_{ab}$. Substitution leaves $2\alpha_{\text{W}}(q_{ab}q^{ab} - q^2)$, the bulk quadratic Weyl density. Formal self-adjointness of the linearized Einstein operator gives the h-equation in (6). The algebraic mass term has no derivative current; polarizing the remaining off-diagonal term gives (7). The final term is the Euler transgression displayed explicitly in Theorem 5.1. □

The parent field f is the target Einstein carrier, while h contains the source Einstein kernel and its metric lifts. Thus, modulo the Euler cut,

$$\boldsymbol{\omega}_{\mathrm{W}}(h_E, h_E') = 0, \qquad \boldsymbol{\omega}_{\mathrm{W}}(h_E, X) = 4\alpha_{\mathrm{W}}\boldsymbol{\omega}_E(h_E, f_X).$$

The mixed block is the ordinary Einstein Green pairing between the source Einstein perturbation and the target carrier. By contrast, the additional–additional entry depends on the choice of metric lift h_X, as the direct calculation found.

Proposition 3.4 (Parent resolvent identity)**.** *Let $\mathcal{E}(\omega)$ be the gauge-fixed linearized Einstein operator and let*

$$(\mathcal{A}f)_{ab} = f_{ab} - g_{ab}f.$$

At every spectral point where $\mathcal{E}(\omega)$ is invertible, the parent Hessian and its inverse are

$$\mathbb{H}(\omega) = 4\alpha_{\mathrm{W}} \begin{pmatrix} 0 & \mathcal{E} \\ \mathcal{E} & -\mathcal{A} \end{pmatrix}, \qquad \mathbb{H}(\omega)^{-1} = \frac{1}{4\alpha_{\mathrm{W}}} \begin{pmatrix} \mathcal{E}^{-1}\mathcal{A}\mathcal{E}^{-1} & \mathcal{E}^{-1} \\ \mathcal{E}^{-1} & 0 \end{pmatrix}.$$

In particular, the metric–metric block is

$$G_{hh} = \frac{1}{4\alpha_{\mathrm{W}}}\,\mathcal{E}^{-1}\mathcal{A}\mathcal{E}^{-1}.$$

Proof. Direct block multiplication on either side gives the identity. No commutation of $\mathcal{A}$ with $\mathcal{E}$ is used. □

This explains the repeated spin-two divisor without cyclic elimination: the metric response is an Einstein resolvent, followed by an algebraic trace-reversal insertion, followed by a second Einstein resolvent. It does not yet provide a physical QNM resolvent, because the required analytic Fredholm realization and its endpoint domains remain open.

Corollary 3.5 (Conditional simple-QNM Laurent coefficient)**.** *Assume such an analytic Fredholm realization, graph-norm boundedness of $\mathcal{A}$, and a simple Einstein resonance ω_n, with*

$$\mathcal{E}(\omega)^{-1} = \frac{P_n}{\omega - \omega_n} + R_n(\omega), \qquad P_n = \frac{u_n \otimes \widetilde{u}_n}{\alpha_n}.$$

Then the coefficient of the double pole in the metric block is

$$\mathrm{Coeff}_{(\omega-\omega_n)^{-2}}\, G_{hh} = \frac{1}{4\alpha_{\mathrm{W}}}\,\frac{\beta_n^{\mathrm{par}}}{\alpha_n^2}\, u_n \otimes \widetilde{u}_n, \qquad \beta_n^{\mathrm{par}} = \langle \widetilde{u}_n, \mathcal{A}u_n\rangle.$$

Proof. The double coefficient is $P_n\mathcal{A}P_n/(4\alpha_{\mathrm{W}})$. Substitution of the rank-one projector gives the displayed formula. □

Projection to the scalar master system is expected to identify β_n^{par} with the radial extension overlap after endpoint and commutator terms are included. That equality is an experiment gate, not an identity silently assumed here.

Remark 3.6 (Parent-overlap and retarded-convolution rails)*.* The first new QNM audit is to evaluate the regularized parent overlap

$$\beta_n^{\mathrm{par}} = \mathrm{FP}\langle \widetilde{f}_n, \mathcal{A}f_n\rangle \text{endpoint terms}$$

on the already certified simple axial QNM disk and compare it with the normalized radial projective overlap. The comparison must include the scalar projection and prove cancellation of the induced commutator terms; agreement is not inferred merely from the common connection Smith type.

Formally, whenever compatible retarded Einstein inverses exist, the metric block has the convolution representation

$$G_{hh}^{\mathrm{ret}}(t)=\frac{1}{4\alpha_{\mathrm{W}}}\int_0^t G_E^{\mathrm{ret}}(t-s)\,\mathcal{A}\,G_E^{\mathrm{ret}}(s)\,\mathrm{d}s.$$

This specifies a second experiment: compare direct six-state evolution with two sequential Einstein evolutions and use the flat-space biwave kernel $\theta(t-r)/(8\pi)$, up to the wave-operator sign convention, as the first control. The algebraic convolution identity does not certify a Schwarzschild retarded propagator, time-domain boundedness, or decay.

The pure-target-gauge part of this carrier has a concrete geometric meaning. If $q_{ab}=\mathcal{L}_\eta g_{ab}$ and $F_{ab}=2\nabla_{[a}\eta_{b]}$, the carrier constraint is precisely

$$\nabla^a F_{ab}=0.$$

A source Weyl transformation $h_{ab}\mapsto h_{ab}+\Phi g_{ab}$ induces $\eta_a\mapsto\eta_a-\nabla_a\Phi/2$, so F is invariant. Thus source Weyl gauge becomes Maxwell gauge of the target Einstein vector. This target gauge is not gauge-trivial as a perturbation of the original Weyl metric.

The action fixes its sign. With $\tau=q^a{}_a$, the Ricci-factorized quadratic density is

$$2\alpha_{\mathrm{W}}\left(q_{ab}q^{ab}-\tau^2\right).$$

For $q=\mathcal{L}_\eta g$,

$$q_{ab}q^{ab}-\tau^2=F_{ab}F^{ab}+4\nabla_a\left(\eta_b\nabla^b\eta^a-\eta^a\nabla_b\eta^b\right).$$

Consequently the spin-one bulk action is

$$S^{(2)}_{\mathrm{spin1}}=2\alpha_{\mathrm{W}}\int F_{ab}F^{ab}=-8\alpha_{\mathrm{W}}S_{\mathrm{Maxwell}}$$

modulo a boundary term. For $\alpha_{\mathrm{W}}>0$, this is the opposite kinetic sign to conventional Maxwell theory.

The static Schwarzschild family is a finite-dimensional shadow of the same nullity. In the residual-basic normalization of the source archive, $\gamma=k=0$ gives $D_2=H=0$, while

$$S_h=64\pi^2\alpha_{\mathrm{W}},\qquad T_\chi=\frac{1}{4\pi},$$

so the restricted first law is the identity $0=T_\chi\,0$. We use this only as a consistency corollary; the general static thermodynamic classification remains in the companion manuscript.

Let the background be Ricci-flat. For a metric perturbation h_{ab}, put

$$\psi_{ab}=\delta R_{ab}[h],\qquad \psi=g^{ab}\psi_{ab}.$$

Linearization of the Bach tensor gives

$$\delta B_{ab}[h]=\frac{1}{2}\Box\psi_{ab}+C_{acbd}\psi^{cd}-\frac{1}{6}\nabla_a\nabla_b\psi-\frac{1}{12}g_{ab}\Box\psi. \tag{8}$$

In the axial sector $\psi=0$, so the carrier equation is a second-order Lichnerowicz-type equation

$$L_{\mathrm{car}}\psi_{ab}:=\frac{1}{2}\Box\psi_{ab}+C_{acbd}\psi^{cd}=0. \tag{9}$$

Proposition 3.7 (Ricci–Bach exact sequence)**.** *On the declared axial Schwarzschild $\ell = 2$ mode class, the linearized Ricci map fits into the exact sequence*

$$0 \longrightarrow \mathcal{E}_{\mathrm{RW}} \longrightarrow \mathcal{B}_{\mathrm{axial}} \xrightarrow{\delta\,\mathrm{Ric}} \mathcal{C}_{\mathrm{axial}} \longrightarrow 0, \tag{10}$$

where $\mathcal{E}_{\mathrm{RW}}$ is the two-dimensional Regge–Wheeler Einstein module and $\mathcal{C}_{\mathrm{axial}}$ is the realized second-order Ricci carrier. The sequence is exact but has no certified canonical metric splitting.

Proof sketch. Equation (8) proves that the Ricci image of a Bach solution obeys (9). The complete reconstruction audit verifies every Ricci row, including the independent $v\varphi$ equation omitted in an earlier shallow reduction. Its zero fibre is exactly the linearized Einstein Regge–Wheeler system. Exact recurrence and reconstruction certificates show that the declared carrier is realized. A rational splitting is separately obstructed by the extension invariants discussed in Section 4. □

3.1 Horizon reach

In ingoing Eddington–Finkelstein coordinates the carrier horizon is a regular singular point. For every nonzero real frequency its ingoing analytic solution space is two-dimensional. Thus horizon regularity alone does not imply $\psi_{ab} = 0$.

Corollary 3.8 (Local horizon nonselection)**.** *Future-horizon analyticity admits nonzero Ricci-carrier data in addition to the Einstein kernel. Hence it does not select the Einstein image within the axial Bach solution space.*

4 The axial filtration and its partial jet

After exact factor reduction the six-state axial system has a filtration

$$0 \subset F_1 \subset F_2 \subset F_3 = M_{\mathrm{axial}},$$

with

$$F_1 \simeq M^{(2)}_{\mathrm{RW}}, \qquad F_2/F_1 \simeq M^{(2)}_{\mathrm{RW}}, \qquad F_3/F_2 \simeq M^{(1)}_{\mathrm{RW}}. \tag{11}$$

Here $M^{(2)}_{\mathrm{RW}}$ is the axial spin-two Regge–Wheeler module and $M^{(1)}_{\mathrm{RW}}$ is the spin-one quotient with potential

$$V_1(r) = \frac{6(r-2)}{r^3}$$

in units $M = 1$.

The filtration now has a covariant associated-graded interpretation:

$$\mathrm{gr}\,\mathcal{H}_{B,\mathrm{axial}} \simeq M^{(h)}_{\mathrm{RW}} \oplus M^{(q)}_{\mathrm{RW}} \oplus M^{(\eta)}_{\mathrm{Maxwell}}.$$

The first factor is the source Einstein kernel, the second is the physical target Einstein carrier, and the third is the Maxwell equation for its target-gauge vector. The certified $\ell = 2$ lift proves this complete realization. The analogous all-ℓ lift remains a prediction, not a theorem of this paper.

In factor variables $X, Y, Z \in \mathbb{C}^2$, the generator is

$$\begin{aligned} X' &= AX + EY + CZ, \\ Y' &= AY + DZ, \\ Z' &= A_1 Z. \end{aligned} \tag{12}$$

The two upper sources share one image line:

$$E = USJ, \qquad C = USN.$$

Consequently (12) is exactly the spin-two-row first jet of the four-state family

$$\mathcal{B}(\tau) = \begin{pmatrix} A + \tau E & D + \tau C \\ 0 & A_1 \end{pmatrix}. \tag{13}$$

Over the dual numbers $\mathbb{C}[\varepsilon]/(\varepsilon^2)$, the six-state solution is simply $(Y + \varepsilon X, Z)$.

Theorem 4.1 (Exact axial partial-jet filtration)**.** *Every interior transport matrix has the form*

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \tag{14}$$

where

$$\begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix}$$

is the transport of (13)*. In particular*

$$\det \Phi_6 = (\det P)^2 \det R.$$

The repeated-spin-two extension is non-split over the generic rational differential field and is not generated, modulo rational gauge, by frequency, Schwarzschild-mass, or angular-eigenvalue differentiation.

The jet identity is useful both conceptually and numerically. A triangular connection

$$T = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & f \end{pmatrix}$$

is the jet of

$$C(\tau) = \begin{pmatrix} a(\tau) & d(\tau) \\ 0 & f \end{pmatrix}, \quad \dot{a} = b, \quad \dot{d} = c.$$

Therefore

$$T^{-1} = \begin{pmatrix} a^{-1} & -b/a^2 & (bd - ac)/(a^2 f) \\ 0 & a^{-1} & -d/(af) \\ 0 & 0 & f^{-1} \end{pmatrix}, \tag{15}$$

and the three apparently independent extension ratios are one base transfer and its intrinsic derivative.

Remark 4.2 (Critical Einstein–Weyl mass jet). The second-order parent action admits the signed squared-mass deformation

$$S_m^{(2)} = 4\alpha_{\mathrm{W}} \int \sqrt{-g} \left[\phi^{ab} \mathcal{E}_{ab}[h] - \frac{1}{2}\left(\phi_{ab}\phi^{ab} - \phi^2\right) + \frac{m}{2} h^{ab} \mathcal{E}_{ab}[h] \right].$$

Its equations are

$$\mathcal{E}h = \mathcal{A}\phi, \qquad \mathcal{E}\phi + m\mathcal{E}h = 0, \qquad (\mathcal{E} + m\mathcal{A})\phi = 0.$$

If $\phi(m)$ is a smooth tensor branch through $m=0$, then

$$\mathcal{E}\dot{\phi}+\mathcal{A}\phi_0=0, \qquad \mathcal{E}(h+\dot{\phi})=0,$$

and hence $[h]=-[\partial_m\phi]_{m=0}$ modulo the Einstein kernel. On the transverse-traceless sector,

$$\{\mathcal{E}(\mathcal{E}+m)\}^{-1}=\frac{1}{m}\{\mathcal{E}^{-1}-(\mathcal{E}+m)^{-1}\}\longrightarrow\mathcal{E}^{-2}.$$

This is the exact covariant critical-mass jet underlying the coalescence of the massless and massive spin-two branches in Einstein–Weyl gravity [10]. At infinity,

$$k=\sqrt{\omega^2-m}, \qquad \partial_m k|_0=-\frac{1}{2\omega},$$

so the moving massive phase produces the generalized $r_* e^{\pm i\omega r_*}$ partner, while the Schwarzschild Coulomb exponent has zero first m-derivative. The finite-mass branch-sign involution is singular in the confluent basis and its renormalized residue is the nilpotent jet map.

What is not yet proved is that the intrinsic radial parameter τ in (13) equals $-m$. That identification requires the exact projective comparison

$$[\mathcal{I}_{\text{mass}}]=[\mathcal{I}_{\text{Bach}}] \quad \text{in} \quad \mathbb{C}(r)/\mathcal{K}_U\mathbb{C}(r).$$

Accordingly no physical massive-QNM slope is inferred here from the intrinsic shear b.

Even equality of these local cocycles would prove only equivalence of the repeated spin-two blocks up to rational triangular gauge and parameter rescaling. The global identity $b=-\partial_m a|_0$ additionally requires the complete mixed Maxwell deformation, compatible moving endpoint phases, and matched Jost-germ normalizations. Identifying β_n/α_n with a massive-QNM slope further requires the analytic Fredholm realization. If the local classes do agree nontrivially, the critical residue of the finite-mass branch-sign operator must be proportional to the unique nilpotent N of Theorem 4.9.

4.1 The spin-one quotient is not a rational spin-two transform

The distinct scalar labels in (11) cannot be removed by a local rational Darboux redefinition.

Proposition 4.3 (No rational spin-one/spin-two intertwiner)**.** *For every fixed real $\omega>0$,*

$$\mathrm{Hom}_{\mathbb{C}(r)[D]}\big(M^{(2)}_{\mathrm{RW}},M^{(1)}_{\mathrm{RW}}\big)=0, \qquad \mathrm{Hom}_{\mathbb{C}(r)[D]}\big(M^{(1)}_{\mathrm{RW}},M^{(2)}_{\mathrm{RW}}\big)=0.$$

Proof. Modulo the source equation, any rational differential map reduces to $P=a(r)D+b(r)$. Requiring it to map all solutions of one scalar factor to the other gives, in either direction,

$$\begin{aligned}0=\mathcal{H}_\omega a&=r^4(r-2)^2a^{(4)}+r^3(r-2)(3r+4)a^{(3)}\\&\quad+r^2(4\omega^2r^4-24r^2+62r-12)a''\\&\quad+r(12\omega^2r^4-90r+60)a'+(90r-60)a.\end{aligned} \tag{16}$$

The local exponents are

$$r=0:\ -1,1,3,5, \qquad r=2:\ 0,1,4i\omega,-4i\omega.$$

A rational solution therefore has at most a simple pole at $r = 0$, no pole at the horizon, and no ordinary-point pole. Dominant balance at infinity permits only $a \sim r^0$ or $a \sim r^{-2}$. Hence $a = A + B/r$. Exact substitution yields

$$\mathcal{H}_\omega\left(A + \frac{B}{r}\right) = 90Ar - 60A - 4B\omega^2 r^3 - 42Br + 220B,$$

which vanishes for real $\omega > 0$ only when $A = B = 0$. The remaining compatibility equation then forces $b = 0$. $\square$

Thus the negative spin-one quotient is neither a disguised spin-two mode nor a rational local spin-changing field redefinition. This is a statement about rational differential modules; it does not exclude nonlocal, meromorphic, or scattering-dependent transforms.

4.2 Real-axis simplicity and the local C obstruction

The non-split extension also constrains possible positive-metric constructions. The statement is local and classical; it is not a theorem about a quantum Hilbert space.

Theorem 4.4 (Real-axis simplicity and scalar endomorphism rings)**.** *Let $\ell \geq 2$, $\Lambda = \ell(\ell+1)$, and*

$$L_{2,\ell} = D^2 + \omega^2 - f\left(\frac{\Lambda}{r^2} - \frac{6}{r^3}\right), \qquad L_{1,\ell} = D^2 + \omega^2 - f\frac{\Lambda}{r^2}, \quad D = f\partial_r.$$

For every real $\omega > 0$, both differential modules are simple over $\mathbb{C}(r)[D]$, and

$$\mathrm{End}_{\mathbb{C}(r)[D]}(M_{2,\ell}) = \mathrm{End}_{\mathbb{C}(r)[D]}(M_{1,\ell}) = \mathbb{C}.$$

The spin-two equation has the exact same-sign algebraically-special controls

$$\omega = \sigma i\,\frac{\Lambda(\Lambda-2)}{12}, \qquad y = e^{\sigma i\omega r_*}\left(1 + \frac{6}{(\Lambda-2)r}\right), \qquad \sigma = \pm 1.$$

For $\ell = 2$, a separate opposite-sign audit excludes $\omega = i/4, i/2, i$ as rational reducibility points.

Proof. Reducibility is equivalent to a rational solution of the Riccati equation

$$Dw + w^2 + \omega^2 - V_{s,\ell} = 0.$$

At $r = 0$, the spin-two exponents are $-1, 3$, the Maxwell exponents are $0, 2$, and at the horizon both equations have exponents $\pm 2i\omega$. A rational prefactor cannot connect opposite horizon and infinity Jost signs for real $\omega \neq 0$, because that would require the noninteger rational power $4i\omega$. Ordinary-point regularity and dominant balance at infinity therefore reduce a possible spin-two hyperexponential solution to

$$y = e^{\sigma i\omega r_*}\left(A + \frac{B}{r}\right).$$

Exact substitution gives

$$L_{2,\ell}y = -\frac{i(r-2)e^{\sigma i\omega r_*}}{r^4}\left[(-i\Lambda A + 2\sigma\omega B)r + i\{6A + (2-\Lambda)B\}\right].$$

A nonzero solution forces

$$B = \frac{6A}{\Lambda - 2}, \qquad \omega = \sigma i\,\frac{\Lambda(\Lambda-2)}{12},$$

which is not positive real. In the Maxwell case the same local analysis leaves only a constant rational prefactor, and its residual $-f\Lambda A/r^2$ is nonzero.

For the endomorphism statement, remove the constant trace. A trace-free horizontal endomorphism is determined by $q \in \mathbb{C}(r)$ satisfying

$$\mathcal{K}_U q = 0, \qquad \mathcal{K}_U = D^3 + 4UD + 2D(U).$$

The spin-two symmetric-square exponents at $r = 0$ are $-2, 2, 6$, so rationality reduces the search to

$$q = q_0 + \frac{q_1}{r} + \frac{q_2}{r^2}.$$

Three exact coefficient consequences are

$$-4\omega^2 q_1 = 0, \qquad 4(\Lambda q_0 - 2\omega^2 q_2) = 0, \qquad -12(\Lambda + 3)q_0 = 0,$$

where the last follows after the preceding eliminations. Hence $q_0 = q_1 = q_2 = 0$. For Maxwell the symmetric-square exponents are $0, 2, 4$, so $q = q_0$, while

$$\mathcal{K}_{U_1}(q_0) = \frac{4\Lambda q_0 (r-3)(r-2)}{r^5};$$

again $q_0 = 0$. The separately certified finite ansätze at the three imaginary frame events have zero kernel for both possible Jost signs. □

Remark 4.5 (Complex reducibility confinement). For complex ω, a rational hyperexponential solution with opposite horizon and infinity Jost signs requires a rational prefactor carrying the horizon exponent difference $4i\omega$. Hence every remaining opposite-sign reducibility candidate obeys

$$4i\omega \in \mathbb{Z}.$$

The same-sign terminal ansatz gives only the static and algebraically special controls. Thus rational reducibility is confined to the quarter-integer imaginary lattice. This is a confinement theorem, not a complete classification: arbitrarily high integral horizon powers remain to be audited. For $\ell = 2$, $i/4, i/2, i$ are already excluded and $\omega = \pm 2i$ are the algebraically special controls.

Proposition 4.6 (No branch-resolving rational involution)**.** *Let*

$$0 \longrightarrow M_{\mathrm{RW}} \longrightarrow E_{\mathrm{Bach}} \longrightarrow M_{\mathrm{RW}} \longrightarrow 0$$

be the generic repeated-spin-two extension. There is no rational differential-module endomorphism C with $C^2 = 1$ that acts by $+1$ on the Einstein submodule and by -1 on the quotient.

Proof. In a triangular frame,

$$\mathcal{A} = \begin{pmatrix} A & E \\ 0 & A \end{pmatrix}.$$

A filtration-compatible involution with the declared actions must be

$$C = \begin{pmatrix} I & Q \\ 0 & -I \end{pmatrix}.$$

Covariant constancy,

$$C' + C\mathcal{A} - \mathcal{A}C = 0,$$

requires

$$Q' + QA - AQ = -2E.$$

Over characteristic zero this says that the extension cocycle is a rational coboundary. It would split the extension, contradicting the certified nonzero rational extension class. □

The scalar classification makes the stronger categorical statement directly applicable on the physical real axis.

Lemma 4.7 (Involutions of a simple nonsplit self-extension)**.** *Over a characteristic-zero differential field, let*

$$0 \longrightarrow M \longrightarrow E \longrightarrow M \longrightarrow 0$$

be a nonsplit self-extension of a simple differential module M. Every differential-module involution of E is $+I$ or $-I$.

Proof. For an involution C, the operators $P_{\pm} = (I \pm C)/2$ are complementary differential-module idempotents. If $C \neq \pm I$, both images are nonzero. Since E has composition length two with identical simple factors, both images have length one and give a direct-sum splitting of the extension, a contradiction. ☐

Corollary 4.8 (Only scalar involutions on the physical axial block)**.** *For axial $\ell = 2$ and every real $\omega > 0$, the only rational differential-module involutions of the repeated-spin-two Bach block are*

$$C = +I, \qquad C = -I.$$

Neither makes its hyperbolic flux form positive.

Proof. The earlier cocycle reduction used a left-null witness proportional to $\omega^2 - 3$, which did not classify that specialization. The same exhaustive coefficient system has the fixed minors

$$\det M_{[0,1,2]} = 3456\omega^{10}, \qquad \det[M \mid \mathrm{rhs}]_{[0,1,2,5]} = -645120 i\omega^9.$$

Thus its coefficient and augmented ranks are respectively three and four for every $\omega > 0$, including $\omega^2 = 3$; the extension is nonsplit throughout the physical real axis. Apply Theorem 4.4 and Lemma 4.7. Multiplication by either remaining scalar sign preserves the inertia of the hyperbolic plane. ☐

Theorem 4.9 (Complete local commutant)**.** *For axial $\ell = 2$ and every real $\omega > 0$, the commutant of the nonsplit repeated-spin-two Bach module is*

$$\mathrm{End}(E_{\mathrm{Bach}}) \simeq \mathbb{C}[\varepsilon]/(\varepsilon^2).$$

More precisely, every endomorphism is uniquely

$$\Phi = aI + bN, \qquad N^2 = 0.$$

Consequently

$$\mathrm{Aut}(E_{\mathrm{Bach}}) = \{aI + bN : a \neq 0\}, \qquad \mathrm{Idem}(E_{\mathrm{Bach}}) = \{0, I\},$$

the only involutions are $\pm I$, every diagonalizable local endomorphism is scalar, and every finite-order local automorphism is scalar.

Proof. The embedded copy of M_{RW} is the unique simple submodule: a second simple submodule mapping nontrivially to the quotient would split the extension. Hence every endomorphism preserves it and induces scalars a, d on the submodule and quotient. Naturality of the nonzero extension class e gives $ae = ed$, so $a = d$. Then $\Phi - aI$ vanishes on the submodule and quotient and factors through the quotient-to-submodule map, giving bN, with $N^2 = 0$. The remaining assertions follow from

$$(aI + bN)^k = a^k I + k a^{k-1} bN$$

in characteristic zero. ☐

Corollary 4.10 (No local positive metric operator)**.** *No rational local dynamically compatible operator η, even without the condition $\eta^2 = I$, makes the repeated-spin-two flux positive definite.*

Proof. In the canonical null basis,

$$G_2 = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad g > 0.$$

Theorem 4.9 gives $\eta = aI + bN$. Krein self-adjointness forces $a, b \in \mathbb{R}$, and

$$G_2\eta = \begin{pmatrix} 0 & ga \\ ga & gb \end{pmatrix}, \qquad \det(G_2\eta) = -g^2a^2.$$

For $a \neq 0$ the form is indefinite; for $a = 0$ it is degenerate. □

The endpoint Krein diagonalization is algebraic, but a bulk dynamical diagonalization would produce nontrivial differential-module projectors and split the extension. The theorem excludes rational local involutions only; nonlocal, spectral, scattering-dependent, and BRST constructions remain open.

The endpoint form supplies a complementary positivity obstruction, stated after its exact Gram is displayed below.

5 Action-derived endpoint flux

The sphere-integrated Lee–Wald current is fixed by the off-shell Green identity

$$\partial_t F^t(h_A, h_B) + \partial_r F^r(h_A, h_B) = 4\alpha_{\mathrm{W}} \int_{S^2} \sqrt{q}\, \big(h_B^{ab}\delta B_{ab}[h_A] - h_A^{ab}\delta B_{ab}[h_B]\big). \tag{17}$$

It is therefore not an arbitrarily normalized Wronskian of a chosen master field.

Theorem 5.1 (Euler cut current and Einstein wave-packet isotropy)**.** *Let $\boldsymbol{\omega}_{C^2}$ be the literal Weyl-action presymplectic current and $\boldsymbol{\omega}_{\mathrm{Ric}}$ the current derived from the Ricci-factorized bulk action. With*

$$\Theta_{E_4}(\delta) = 2\alpha_{\mathrm{W}}\epsilon_{abcd}\,\delta\Gamma^{ab} \wedge R^{cd}, \qquad k_{E_4}(\delta_1, \delta_2) = 2\alpha_{\mathrm{W}}\epsilon_{abcd}\,\delta_1\Gamma^{ab} \wedge \delta_2\Gamma^{cd},$$

one has

$$\boldsymbol{\omega}_{C^2} - \boldsymbol{\omega}_{\mathrm{Ric}} = dk_{E_4}.$$

On axial Einstein modes, restoring the Fourier factor $e^{i(\omega_1+\omega_2)t}$, the exact radial current is

$$F^r_{EE} = -\frac{192\pi\alpha_{\mathrm{W}}(\omega_1^2 - \omega_2^2)}{5\omega_1\omega_2 r}\,\Psi_1\Psi_2 e^{i(\omega_1+\omega_2)t} = \partial_t Q_{EE},$$

where

$$Q_{EE} = \frac{192\pi i\alpha_{\mathrm{W}}(\omega_1 - \omega_2)}{5\omega_1\omega_2 r}\,\Psi_1\Psi_2 e^{i(\omega_1+\omega_2)t}.$$

Therefore every pair of smooth compact-frequency Einstein wave packets with support bounded away from $\omega = 0$ has zero integrated Einstein–Einstein flux through each finite-radius cylinder. The Einstein wave-packet subspace is totally isotropic. The same conclusion holds at a null or horizon endpoint whenever the declared packet bounds justify interchanging the endpoint limit with the time integral.

Proof. The Chern–Weil variation $\delta R^{ab} = D\delta\Gamma^{ab}$, metric compatibility, and graded antisymmetry give $\boldsymbol{\omega}_{E_4} = dk_{E_4}$. The displayed Fourier identity follows from $i(\omega_1+\omega_2)i(\omega_1-\omega_2) = -(\omega_1^2-\omega_2^2)$. On the declared packet core the inverse Fourier transforms and the cut bilinear are Schwartz in t; hence $\int_{\mathbb{R}} F^r_{EE}\mathrm{d}t = Q_{EE}(+\infty) - Q_{EE}(-\infty) = 0$. □

The theorem does not set the literal monochromatic current to zero pointwise. It identifies that current as an Euler cut contribution. The mixed Einstein–additional block is different: the exact all-$\omega > 0$ endpoint Gram below forces it to be nonzero, while the interior fixtures are retained only as independent checks. This cross block is not established to be Euler-exact. Thus the Einstein wave-packet space is isotropic but nonradical in the full Weyl phase space. Theorem 3.3 identifies its non-Euler part with the ordinary Einstein Green pairing.

Let $G_-(\omega)$ be the incoming null-infinity Gram in the three-dimensional factor trace space. Define

$$E = EI0, \qquad R = XI0 - \frac{i}{\omega}XI1,$$

and let S be the unique G_--orthogonal lift of the unit spin-one quotient. After the harmless Einstein shear making the second spin-two factor null, the exact factor Gram is

$$\widehat{G}_-^{\mathrm{fac}} := \frac{G_-}{\pi\alpha_{\mathrm{W}}} = \begin{pmatrix} 0 & \dfrac{576}{5}\omega & 0 \\ \dfrac{576}{5}\omega & 0 & 0 \\ 0 & 0 & -\dfrac{32}{15\omega} \end{pmatrix}. \tag{18}$$

The availability of this null lift is structural. Any Hermitian block

$$\begin{pmatrix} 0 & a \\ \bar{a} & b \end{pmatrix}, \qquad a \neq 0,$$

changes under $X \mapsto X + cE$ by $b \mapsto b + c\bar{a} + \bar{c}a$. A choice of c always sets the second diagonal entry to zero, while the determinant remains $-|a|^2$. Accordingly the parent action predicts a canonical off-diagonal source/target spin-two kinetic block; the wrong-sign Maxwell quotient then supplies the separate negative line. Direct raw-Weyl current reduction remains an independent audit of

$$G_{\text{raw Weyl}} - G_{\text{parent}} = G_{\text{Euler cut}}.$$

Theorem 5.2 (Canonical incoming Krein anatomy)**.** *For every real $\omega > 0$,*

$$\operatorname{inertia} G_-(\omega) = (1, 2, 0).$$

The repeated spin-two extension is a hyperbolic plane and the intrinsic spin-one quotient metric is strictly negative:

$$g_1 = -\pi\alpha_{\mathrm{W}}\frac{32}{15\omega}.$$

In normalized channels

$$P_2 = \frac{E+R}{\sqrt{2a}}, \qquad N_2 = \frac{E-R}{\sqrt{2a}}, \qquad N_1 = \frac{S}{\sqrt{b}},$$

where $a = 576\omega/5$ and $b = 32/(15\omega)$, the Gram is

$$\operatorname{diag}(1, -1, -1).$$

Proposition 5.3 (Einstein-line positivity obstruction)**.** *Let W be any subspace of an endpoint trace space that contains the nonzero Einstein line $\mathbb{C}E$, with $G(E,E)=0$. If $G|_W$ is nondegenerate, then it is indefinite. Equivalently, every semidefinite restriction retaining $\mathbb{C}E$ is degenerate.*

Proof. If $G(E,w)=0$ for every $w\in W$, then E lies in the radical of $G|_W$. Otherwise choose $x\in W$ with $a=G(E,x)\neq 0$. On $\operatorname{span}\{E,x\}$ the Gram is

$$\begin{pmatrix} 0 & a \\ \bar{a} & b \end{pmatrix}, \qquad \det = -|a|^2 < 0.$$

That plane has inertia $(1,1,0)$, independently of b. □

Corollary 5.4 (Positive-graph and cotangent obstructions)**.** *In the normalized endpoint space $J=\operatorname{diag}(1,-1,-1)$, no closed uniformly positive subspace contains the pure Einstein channel. A nonnegative subspace can contain it only as a null direction. Moreover, if B_2 is the two-dimensional spin-two trace space and $E\subset B_2$ the Einstein line, then the flux pairing induces the canonical isomorphism*

$$B_2/E \cong E^*.$$

Proof. A maximal uniformly positive line is the graph of $K:\mathbb{C}\to\mathbb{C}^2$ with $\|K\|<1$. In the signature basis the Einstein vector is $(1,1,0)/\sqrt{2}$, whose inclusion requires $K(1)=(1,0)$ and hence $\|K\|=1$. For the second statement, the spin-two plane is nondegenerate and hyperbolic, so $E^\perp = E$. The map

$$[x] \longmapsto \big(e \mapsto G(e,x)\big)$$

is therefore a well-defined nondegenerate duality from B_2/E to E^*. □

For wave packets the global graph statement uses an operator contraction between the positive and negative Hilbert summands. The pointwise $K(\omega)$ formulation applies to decomposable, frequency-multiplication-invariant subspaces. The endpoint duality is canonical, but choosing a metric lift is only an affine splitting and does not split the radial differential module.

Thus the inherited pure-Weyl form presents a trilemma. Retaining only the Einstein directions gives a null, degenerate sector. Adding a symplectic partner makes the spin-two plane nondegenerate but indefinite. Positivity requires a modified inner product. In particular, if a fundamental symmetry preserved the Einstein line, $CE=\pm E$, then $G(E,CE)=0$; any positive fundamental symmetry must mix the two null spin-two directions. The elementary endpoint exchange is a classical Krein fundamental symmetry, not Mannheim's dynamical C. Together with Corollary 4.8, this shows that any successful positive dynamical construction cannot be a rational local involution. A nonlocal, spectral, scattering-dependent, or BRST-level construction is not excluded.

Theorem 5.5 (Spectral fundamental symmetry on compact bands)**.** *For every real $\omega>0$, define*

$$C_-(\omega) = \operatorname{sgn} G_-(\omega) = G_-(\omega)\{G_-(\omega)^2\}^{-1/2}.$$

Then

$$C_-^2 = I, \qquad C_-^\dagger G_- = G_- C_-, \qquad G_- C_- = |G_-| > 0.$$

Its pullback to the horizon-regular solution fiber,

$$C_{\mathrm{sol}} = T_-^{-1} C_- T_-,$$

is a positive fundamental symmetry for $G_{\mathrm{sol}} = T_-^\dagger G_- T_-$. On every compact $I \Subset (0,\infty)$, it defines a bounded involution on $L^2(I;\mathbb{C}^3)$, and its positive norm is uniformly equivalent to the coefficient norm.

Proof. The first three identities are the Hermitian functional calculus for an invertible Hermitian matrix. Pullback gives

$$G_{\mathrm{sol}} C_{\mathrm{sol}} = T_-^\dagger |G_-| T_- > 0.$$

Continuity, compactness, and invertibility of G_- and T_- give the uniform band bounds. □

This proves existence of a nonlocal spectral C, rather than a local branch observable. It does not make C_{sol} canonical, covariant, causal, holomorphic in complex frequency, or BRST compatible. In particular, matrix sign uses the auxiliary positive coefficient inner product: for a nonunitary frame change $G \mapsto M^\dagger G M$, generally

$$\operatorname{sgn}(M^\dagger G M) \neq M^{-1} \operatorname{sgn}(G) M.$$

For example, $G = \operatorname{diag}(1, -1)$ and $M = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ make the right-hand side non-Hermitian, whereas the left-hand side is Hermitian.

The positive Hilbert majorant associated with this Krein form is

$$\|(e, r, s)\|_{\mathrm{maj}}^2 = \int_0^\infty \left[\frac{576}{5} \omega (|e|^2 + |r|^2) + \frac{32}{15\omega} |s|^2 \right] \mathrm{d}\omega. \tag{19}$$

Thus the natural time-domain exponents are

$$(e, r) \in \dot{H}^{1/2}, \qquad s \in \dot{H}^{-1/2}.$$

This identifies the canonical completion but does not itself prove global boundedness of the scattering multiplier.

In the unrescaled incoming factor-Witt basis, after the exact second-null lift, the same construction reads

$$G_-^{\mathrm{fac}} = \frac{384}{5} \begin{pmatrix} 0 & \omega & 0 \\ \omega & 0 & 0 \\ 0 & 0 & -\omega^3 \end{pmatrix}, \qquad C_-^{\mathrm{fac}} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Hence

$$G_-^{\mathrm{fac}} C_-^{\mathrm{fac}} = \frac{384}{5} \operatorname{diag}(\omega, \omega, \omega^3).$$

The natural factor-coefficient completion at threshold has norm

$$\int_0^\epsilon \left[\omega (|u_E|^2 + |u_X|^2) + \omega^3 |u_1|^2 \right] \mathrm{d}\omega.$$

Equation (19) uses the separately unit-normalized quotient coefficient and is related by a frequency-dependent factor-basis change. Neither presentation is uniformly equivalent to unweighted L^2 at $\omega = 0$. Equivalently, the infrared-regular coefficients are

$$\widehat{u}_E = \omega^{1/2} u_E, \qquad \widehat{u}_X = \omega^{1/2} u_X, \qquad \widehat{u}_1 = \omega^{3/2} u_1.$$

Transfer matrices should be normalized as

$$\widehat{T} = W_{\mathrm{out}}^{1/2} T W_{\mathrm{in}}^{-1/2}, \qquad W^{1/2} = \operatorname{diag}(\omega^{1/2}, \omega^{1/2}, \omega^{3/2});$$

this becomes an ordinary conjugation only when the endpoint weights agree.

6 Every incoming channel is populated

Let $T_-(\omega)$ map future-horizon-regular coefficients to incoming null-infinity factor traces. Its factor form is triangular:

$$T_- = \begin{pmatrix} a_2 & b & c \\ 0 & a_2 & d \\ 0 & 0 & a_1 \end{pmatrix}, \qquad a_s = A_{\mathrm{in},s}. \tag{20}$$

Lemma 6.1 (Boundary dévissage)**.** *Suppose a filtered differential system has boundary-compatible inclusion and quotient maps, and none of its scalar quotient problems admits a nonzero solution in the declared boundary class. Then the full filtered system has no nonzero solution in that class.*

Proof. Project a full solution to the top quotient. The quotient boundary problem forces the projection to vanish, so the solution lies in the preceding submodule. Repeat through the filtration. The last scalar factor then forces the remaining solution to vanish. No diagonalization or splitting of the extension is required. □

For either scalar Regge–Wheeler factor, real-frequency current conservation gives

$$|A_{\mathrm{in},s}|^2 - |A_{\mathrm{out},s}|^2 = 1. \tag{21}$$

Thus $A_{\mathrm{in},s} \neq 0$ for every real $\omega > 0$.

Theorem 6.2 (Global incoming population)**.** *For the complete axial $\ell = 2$ system,*

$$T_-(\omega) \in GL(3,\mathbb{C}) \qquad \textit{for every real } \omega > 0.$$

Consequently every positive, negative, and null vector of the incoming Krein space is the trace of a unique future-horizon-regular solution.

Proof. A vector in $\ker T_-$ would be horizon regular and purely outgoing at infinity. Its spin-one projection vanishes by (21). Boundary dévissage then removes the carrier and metric spin-two factors successively. Hence the kernel is zero. Since T_- is a 3×3 map, it is invertible. □

The conclusion is stronger than existence of formal negative directions: the entire indefinite incoming space is physically reached within the declared exterior boundary problem.

7 Outgoing population and one-sided Krein scattering

Let T_+ be the future-null-infinity outgoing connection and $H_{\mathcal{H}^+}$ the outward-oriented future-horizon Gram. Conservation of (17) gives

$$H_{\mathcal{H}^+} + T_+^\dagger G_+ T_+ - T_-^\dagger G_- T_- = 0. \tag{22}$$

The explicit entries of T_+ are not needed to decide population on the certified cell. The scalar outgoing factors and the exact filtration give

$$\det T_+ = \rho_+ A_{\mathrm{out},2}^2 A_{\mathrm{out},1}, \qquad \rho_+ \neq 0.$$

Theorem 7.1 (Certified outgoing cell)**.** *For $M = 1$, axial $\ell = 2$, and*

$$I_0 = [0.49995, 0.50005],$$

validated endpoint calculations prove

$$|A_{\mathrm{out},2}| > 0.011326240435330133, \qquad |A_{\mathrm{out},1}| > 0.14680548488890086.$$

Hence $T_+(\omega)$ is invertible throughout I_0. On the exact common cell

$$I_* = [1/2, 10001/20000],$$

the incoming, outgoing, and future-horizon coefficient spaces are nondegenerate three-channel Krein spaces of inertia $(1, 2, 0)$.

Theorem 7.2 (Global one-ended radiative nonselection)**.** *For $M = 1$, axial $\ell = 2$:*

1. *for every real $\omega > 0$, there is a global future-horizon-regular exterior Bach solution with finite incoming null-infinity trace and $\mathfrak{q}[h] \neq 0$;*
2. *for every $\omega \in I_*$, there is such a solution with any prescribed nonzero outgoing Maxwell-quotient trace, and again $\mathfrak{q}[h] \neq 0$.*

Consequently the standard one-ended radiative conditions of Definition 1.1 do not imply the linearized Einstein constraint, even modulo source diffeomorphism and Weyl gauge.

Proof. For the first statement choose a nonzero incoming Maxwell factor trace. Theorem 6.2 gives a unique future-horizon-regular exterior solution realizing it. The factor inclusion and quotient maps preserve the declared endpoint classes. If $\mathfrak{q}[h]$ vanished, its Maxwell two-form and hence its Maxwell quotient trace would vanish, a contradiction. For the second statement use Theorem 7.1 to invert T_+ on I_*, and apply the same gauge-invariant quotient argument. No simultaneous pure-outgoing condition is used. □

In incoming coordinates define

$$R = T_+ T_-^{-1}, \qquad A = T_-^{-1}.$$

Multiplying (22) by $T_-^{-\dagger}$ and T_-^{-1} gives

$$G_- = R^\dagger G_+ R + A^\dagger H_{\mathcal{H}^+} A. \tag{23}$$

Equivalently,

$$\mathscr{S} = \begin{pmatrix} R \\ A \end{pmatrix}$$

is a pseudo-isometric embedding

$$(\mathbb{C}^3, G_-) \hookrightarrow (\mathbb{C}^3 \oplus \mathbb{C}^3, G_+ \oplus H_{\mathcal{H}^+}).$$

Proposition 7.3 (Exact positive-metric scattering test)**.** *Let*

$$J_{\mathrm{out}} = G_+ \oplus H_{\mathcal{H}^+}, \qquad C_{\mathrm{out}} = C_+ \oplus C_H,$$

where C_-, C_+, C_H are Krein-self-adjoint involutions with positive majorants. Under (23), the positive-metric identity

$$\mathscr{S}^\dagger J_{\mathrm{out}} C_{\mathrm{out}} \mathscr{S} = G_- C_-$$

is equivalent to

$$C_{\mathrm{out}} \mathscr{S} = \mathscr{S} C_-.$$

Proof. The intertwining equation and (23) immediately imply the positive identity. Conversely set $\Delta_C = C_{\mathrm{out}}\mathscr{S} - \mathscr{S}C_-$. Expanding $\Delta_C^\dagger J_{\mathrm{out}} C_{\mathrm{out}} \Delta_C$, the two norm terms and the two cross terms reduce, using the Krein and positive identities, to

$$G_-C_- + G_-C_- - G_-C_- - G_-C_- = 0.$$

Positivity of $J_{\mathrm{out}}C_{\mathrm{out}}$ forces $\Delta_C = 0$. □

Thus explicit T_+ supplies a sharp finite-dimensional test of whether independently selected endpoint fundamental symmetries are dynamically compatible. A transported symmetry always exists on $\operatorname{ran}\mathscr{S}$, but it need not split as independent operators on $\mathscr{I}^+$ and $\mathcal{H}^+$.

Proposition 7.4 (Combined-future existence and factorization problem)**.** *For every incoming fundamental symmetry C_-, there exists a fundamental symmetry C_{out} on the combined future space satisfying*

$$C_{\mathrm{out}}\mathscr{S} = \mathscr{S}C_-.$$

The nontrivial physical question is whether one may choose $C_{\mathrm{out}} = C_+ \oplus C_H$.

Proof. The range of the injective Krein isometry $\mathscr{S}$ is nondegenerate. Transport C_- to it by $C_{\mathrm{ran}}\mathscr{S}x = \mathscr{S}C_-x$. The combined future space splits as the Krein-orthogonal direct sum of this range and its orthogonal complement. Choose any fundamental symmetry on the complement and take the direct sum. Factorization over the two future boundaries is additional structure and is exactly the test in Proposition 7.3. □

Corollary 7.5 (Band-limited pseudo-isometry)**.** *Multiplication by T_+ is a bounded isomorphism on $L^2(I_*;\mathbb{C}^3)$. Equation (23) therefore integrates to an exact pseudo-isometry of the complete band-limited wave-packet spaces.*

Corollary 7.6 (Generic positive-real outgoing population)**.** *Assume the factor-normalized scalar Jost coefficients are holomorphic away from the separate threshold $\omega = 0$. Their strict nonvanishing on I_0 proves that neither is identically zero. Therefore the positive-real exceptional set is locally finite and*

$$T_+(\omega) \in GL(3,\mathbb{C})$$

on an open dense, full-measure subset of $(0,\infty)$. On every compact positive-frequency band the corresponding multiplication operator is injective with dense range. It is a bounded isomorphism exactly when the band contains no exceptional frequency.

This does not prove that the exceptional set is empty. The next scattering experiment is to compute the typed matrix T_+, the reflection map R, and their extension shears on a substantial real-frequency band using the partial-jet system (13).

8 Exact threshold structure

The threshold $\omega = 0$ is excluded from the preceding holomorphic genericity argument and must be treated separately. Several exact facts can nevertheless be established before a uniform low-frequency scattering estimate.

For $s \in \{1,2\}$ and integer $\ell \geq 2$, let

$$D = \frac{r-2}{r}\partial_r, \qquad V_{s\ell} = \left(1 - \frac{2}{r}\right)\left[\frac{\ell(\ell+1)}{r^2} + \frac{2(1-s^2)}{r^3}\right],$$

and consider $(D^2 - V_{s\ell})\phi = 0$.

Proposition 8.1 (Exact all-ℓ scalar threshold nonresonance)**.** *The horizon-normalized regular zero mode is*

$$\phi^{(0)}_{s\ell}(r) = \frac{(2\ell)!}{2^{\ell+1}(\ell-s)!(\ell+s)!} r^{\ell+1}\, {}_2F_1\left(s-\ell, -s-\ell; -2\ell; \frac{2}{r}\right).$$

The series terminates, $\phi^{(0)}_{s\ell}(2) = 1$*, and*

$$\phi^{(0)}_{s\ell}(r) \sim \frac{(2\ell)!}{2^{\ell+1}(\ell-s)!(\ell+s)!}\, r^{\ell+1}.$$

A reduction-of-order companion can be normalized to decay as $r^{-\ell}$ *at infinity, but it is logarithmically singular at the horizon. Hence neither scalar factor has a zero-energy resonance for any* $\ell \geq 2$*. For* $\ell = 2$*, the regular modes reduce to*

$$\phi_{0,2}(r) = \frac{r^3}{8}, \qquad \phi_{0,1}(r) = \frac{r^2(2r-3)}{4}.$$

Their explicit independent companions are, up to nonzero constant multiples,

$$\widetilde{\phi}_{0,2} = \frac{r^3}{32}\log\frac{r-2}{r} + \frac{r^2}{16} + \frac{r}{16} + \frac{1}{12} + \frac{1}{8r},$$

and

$$\widetilde{\phi}_{0,1} = \frac{3r^2(2r-3)\log\frac{r-2}{r} + 12r^2 - 6r - 2}{24}.$$

The $\ell = 2$ *regular modes grow as nonzero multiples of* r^3*. The decaying companions behave respectively as* $-1/(5r^2) + O(r^{-3})$ *and* $-1/(10r^2) + O(r^{-3})$*.*

Proof. With $z = 2/r$, substitution reduces the static equation to the Gauss hypergeometric equation with

$$a = s-\ell, \qquad b = -s-\ell, \qquad c = -2\ell.$$

Since a is a nonpositive integer the series terminates. Chu–Vandermonde gives the stated horizon normalization. The second solution is

$$\widetilde{\phi}_{s\ell} = \phi^{(0)}_{s\ell} \int \frac{\mathrm{d}r}{(1-2/r)(\phi^{(0)}_{s\ell})^2}.$$

Because $\phi^{(0)}_{s\ell}(2) = 1$, it contains $\log(r-2)$. Choosing the integration constant from infinity gives $r^{-\ell}$, while the regular solution grows as $r^{\ell+1}$. This exhausts the scalar solution space. Direct exact substitution recovers the displayed $\ell = 2$ controls. $\square$

The exact reduced projective cocycle also exposes the relation between the three factors:

$$\mathcal{I}_{\text{red}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega} = \frac{2i}{5\omega}(V_1 - V_2) + \frac{i\omega(r-2)(2r+3)}{5r^4}. \tag{24}$$

Thus its singular threshold term is exactly the difference of the spin-one and spin-two potentials. On $\phi_{0,2}$ the leading source is

$$\frac{3i}{10\omega}\frac{r-2}{r}.$$

It has the elementary spin-two primitive

$$(D^2 - V_2)p = \frac{r-2}{r}, \qquad p = -\frac{r^2}{4} - \frac{r}{4} - \frac{1}{3} - \frac{1}{2r}. \tag{25}$$

The horizon-fixed choice

$$p_H = p + \frac{25}{12}\phi_{0,2}$$

satisfies $p_H(2) = 0$.

Remark 8.2 (Open matching estimate). Formal two-region matching to the far spherical equation predicts, for all $\ell \geq 2$,

$$|A_{\mathrm{in},s\ell}| \sim |A_{\mathrm{out},s\ell}| \sim \kappa_{s\ell}\omega^{-(\ell+1)}, \qquad \kappa_{s\ell} = \frac{(2\ell)!(2\ell+1)!!}{2^{\ell+2}(\ell-s)!(\ell+s)!},$$

and

$$\Gamma_{s\ell} \sim \kappa_{s\ell}^{-2}\omega^{2\ell+2}.$$

For $\ell = 2$, this gives

$$|A_{\mathrm{in},2}| \sim |A_{\mathrm{out},2}| \sim \frac{15}{16\omega^3}, \qquad |A_{\mathrm{in},1}| \sim |A_{\mathrm{out},1}| \sim \frac{15}{4\omega^3},$$

and therefore

$$|\det T_+| \sim \frac{3375}{1024}\omega^{-9}.$$

The primitive (25) further suggests $b/a^2 = O(\omega^2)$ in the compatible intrinsic normalization. These asymptotics are not used as theorems here. A two-region Volterra remainder controlling the Schwarzschild tail and an endpoint-compatible extension normalization are still required before one may infer a punctured zero-frequency interval of outgoing invertibility or a weighted multiplier bound.

9 Separated-mode stability

In the Fourier convention used here, exponentially growing modes lie in the lower half ω-plane. For the scalar spin-one and spin-two Regge–Wheeler equations, an ingoing-horizon/outgoing-infinity solution in that half-plane decays at both ends. Their real, nonnegative potentials give the standard energy identity, forcing the scalar solution to vanish.

Theorem 9.1 (No growing axial separated mode)**.** *The complete axial $\ell = 2$ six-state system has no nonzero separated solution satisfying the future-horizon-regular and pure-outgoing-infinity conditions in the growth half-plane.*

Proof. The boundary maps preserve the declared complex-frequency classes, including regular replacements at the apparent frame events. Apply Lemma 6.1: the spin-one projection vanishes by the scalar energy identity, then the carrier spin-two quotient vanishes, and finally the metric spin-two factor vanishes. □

This is separated-mode stability. It does not supply limiting absorption, uniform resolvent bounds, threshold control, decay, or bounded time evolution. The extension ratios in (15) can still generate non-normal amplification.

10 A certified intrinsic connection exceptional point

At a simple spin-two quasinormal frequency ω_n, assume the spin-one factor is a local unit. Analytic elimination of that unit reduces the repeated block to

$$T_2(\omega) = \begin{pmatrix} a(\omega) & b(\omega) \\ 0 & a(\omega) \end{pmatrix}, \qquad a(\omega_n) = 0, \quad a'(\omega_n) \neq 0. \tag{26}$$

Proposition 10.1 (Local Smith dichotomy). *If $b(\omega_n) \neq 0$, the full 3×3 connection has Smith valuations*

$$(0, 0, 2),$$

rank two at the QNM, geometric multiplicity one, and one length-two root chain. If $b(\omega_n) = 0$, the valuations are

$$(0, 1, 1),$$

the rank is one, and there are two independent root vectors.

Proof. Work in the local discrete valuation ring with $z = \omega - \omega_n$. In the first case b is a unit, so the ideal of 2×2 minors is the unit ideal while $\det T$ has valuation two. The invariant factors are therefore $(0, 0, 2)$. In the second case $b = ag$, every 2×2 minor is divisible by a, while the minor af has valuation one; hence $(0, 1, 1)$. □

The determinant count is simpler than the local Smith classification. Let D_2 and D_1 be scalar spin-two and Maxwell Evans functions in a domain where the endpoint normalizers are analytic units. Triangularity gives

$$D_B(\omega) = u(\omega) D_2(\omega)^2 D_1(\omega), \qquad u(\omega) \neq 0. \tag{27}$$

Hence every zero-free contour Γ obeys the exact algebraic count

$$N_B(\Gamma) = 2N_2(\Gamma) + N_1(\Gamma). \tag{28}$$

The extension can change geometric multiplicity and Smith type, but not the total algebraic divisor. Thus scalar argument principles suffice for QNM counting; the extension selector is needed only to separate semisimple and defective doubled spin-two roots.

The selector can be computed projectively. Transport the horizon-regular and infinity-outgoing scalar lines to a common matching section and use a Riccati chart $v = Y_2/Y_1$. With

$$\delta(\omega, \tau) = v_\infty(\omega, \tau) - v_H(\omega, \tau),$$

the scalar Evans coefficient has the form $a = g\delta$, where g is an analytic unit. At a simple QNM,

$$\frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n)}{\delta_\omega(\omega_n)} = -\omega_n'(0). \tag{29}$$

Theorem 10.2 (Certified axial connection EP2). *There is a radius-10^{-7} disk centered at*

$$\omega_c = -0.3736716844180418 + 0.0889623156889357\, i$$

that contains exactly one simple spin-two quasinormal zero in the declared $\mathrm{e}^{+i\omega t}$ convention. On that disk:

1. the validated projective Evans contour has winding $+1$*;*

2. δ_ω *excludes zero;*

3. the intrinsic tangent satisfies

$$\delta_\tau \in [0.0010932 \pm 7.60 \times 10^{-8}] + i[-0.0002195 \pm 8.89 \times 10^{-8}],$$

and therefore excludes zero;

4. the spin-one projective mismatch has modulus greater than $1/2500$.

Consequently the complete axial connection has Smith valuations $(0, 0, 2)$*, algebraic multiplicity two, geometric multiplicity one, and one length-two connection root chain at the unique enclosed QNM.*

The exceptional point is intrinsic: the rational self-extension is not a frequency, mass, or angular-eigenvalue tangent modulo admissible rational gauge. A generic non-split *radial* differential-module extension is not, by itself, a Jordan block of the time-translation generator and does not imply a $te^{i\omega t}$ term. Such a temporal interpretation requires a discrete spectral obstruction. Theorem 10.2 supplies that obstruction at one physical QNM at the connection level.

What is certified there is a pole of order two of the inverse *connection* block. To infer the rank-one principal part

$$-\frac{\beta_n}{\alpha_n^2}\frac{u_n \otimes \widetilde{u}_n}{(\omega-\omega_n)^2}$$

for the physical differential resolvent, one must still construct an analytic Fredholm realization with the QNM endpoint conditions and prove graph-norm boundedness of the extension.

11 Polar endpoint reach

The polar carrier includes trace terms:

$$\delta B_{ab} = \frac{1}{2}\Box\psi_{ab} + C_{acbd}\psi^{cd} - \frac{1}{6}\nabla_a\nabla_b\psi - \frac{1}{12}g_{ab}\Box\psi. \tag{30}$$

On the Bianchi-constrained polar $\ell = 2$ system the operator is traceless and has a one-function linearized conformal-gauge direction. On a traceless slice the horizon residue analysis yields, after quotienting the regular conformal direction, a two-parameter physical ingoing-analytic carrier family.

The source and target physical Einstein classes in even parity are governed by the standard gauge-invariant Zerilli–Moncrief scalar [6]. The earlier failure to find an ω-independent Schrödinger scalar within one rational ansatz is therefore only an ansatz failure, not a missing physical master equation. The covariant detour interpretation predicts the polar associated graded

$$M^{(h)}_{\mathrm{Zerilli}} \oplus M^{(q)}_{\mathrm{Zerilli}} \oplus M^{(\eta)}_{\mathrm{Maxwell}}.$$

The Regge–Wheeler and Zerilli scalar equations are related by the Chandrasekhar–Darboux transformation [7]. What remains to be certified is the complete polar image/lift, endpoint units, extension shears, and action Gram; the displayed polar associated graded and its expected inertia are not promoted here to a global polar theorem.

Corollary 11.1 (Parity-complete local endpoint nonselection)**.** *At the linear* $\ell = 2$ *mode level, both axial and polar Ricci carriers have nonzero future-horizon-regular data and propagate on the Einstein characteristics with admissible leading outer falloff. The tested local endpoint diagnostics therefore fail to select the Einstein image in either parity. The global populated Krein and connection theorems of this paper are axial only.*

12 Logical map and evidence states

The title theorem has the following closed dependency chain. Here *fail closed* means that missing data, a timeout, an unsupported domain, or an unverifiable claim is rejected rather than counted as a partial pass. The unfinished explicit T_+, Fredholm, and retarded-convolution calculations are absent from this chain.

Statement	Mathematical dependency and primary reproducibility record	Status and scope	Required for title theorem
Gauge-invariant non-Einstein discriminator	Schouten carrier and Maxwell quotient; `black_hole_programme/phase4/covariant_einstein_maxwell_carrier_v1/certificate.json`	Exact symbolic; Ricci-flat backgrounds, with source diffeomorphism/Weyl quotient stated above	Yes
Axial filtered carrier with a nonzero Maxwell quotient	RW/RW/Maxwell filtration; `black_hole_programme/phase3/axial_rw_lx_triangular_preflight/certificate.json`	Exact symbolic; Schwarzschild, axial $\ell = 2$	Yes
Global future-horizon-regular realization of every incoming trace	Scalar Wronskians, boundary dévissage, and T_- invertibility; `black_hole_programme/phase3/axial_incoming_extended_domain_audit/certificate.json`	Analytic plus computer-assisted; every real $\omega > 0$	Yes
Outgoing realization on I_*	Triangular determinant and certified nonzero scalar outgoing factors; `black_hole_programme/phase3/axial_outgoing_population_cell_half_v1/certificate.json`	Validated interval theorem; $M = 1$, axial $\ell = 2$, $I_* = [1/2, 10001/20000]$	Yes, for the outgoing clause only
Explicit T_+ entries, QNM Fredholm pole, retarded convolution	Separate open computations	Not established and not imported	No

The theorem statements and proofs are the mathematical authority. The machine-readable records are proof objects where a computer-assisted lemma is used and reproducibility evidence elsewhere; rerunning a producer is reproduction, not independent verification.

The core structure is

$$
\begin{array}{ccccccccc}
0 & \longrightarrow & M_{\mathrm{RW}}^{(2)} & \longrightarrow & M_{\mathrm{Bach}}^{\mathrm{axial}} & \xrightarrow{\delta\,\mathrm{Ric}} & M_{\mathrm{car}} & \longrightarrow & 0 \\
 & & \cup & & & & \cap & & \\
 & & F_1 & \subset & F_2 & \subset & F_3 = M_{\mathrm{axial}}, & &
\end{array}
$$

with successive factors $(2, 2, 1)$. For real frequency, horizon data $h \in \mathbb{C}^3$ maps to

$$
\begin{tikzcd}
 & h \arrow[dl, "T_-"'] \arrow[dr, "T_+"] & \\
\mathscr{I}^-_{\mathrm{in}} & & \mathscr{I}^+_{\mathrm{out}},
\end{tikzcd}
$$

and the action current supplies

$$H_{\mathcal{H}^+} + T_+^\dagger G_+ T_+ - T_-^\dagger G_- T_- = 0.$$

Statement	Evidence state	Boundary
Ricci-factorized Hessian and null Einstein kernel	Exact symbolic theorem	Every four-dimensional Ricci-flat background, modulo Euler boundary/corner terms
Schouten–Einstein carrier and wrong-sign Maxwell target gauge	Exact symbolic theorem with independent certificate	Every four-dimensional Ricci-flat background; boundary divergence retained

Statement	Evidence state	Boundary
Second-order parent system and factorized Einstein current	Exact symbolic theorem with independent certificate	Every four-dimensional Ricci-flat background modulo Euler boundary/corner terms
Parent resolvent identity and rank-one Laurent algebra	Exact algebra; QNM specialization conditional	Gauge-fixed invertible Einstein operator; a physical pole additionally requires the analytic Fredholm and bounded-insertion hypotheses
Critical Einstein–Weyl parent mass jet and confluent branch-sign limit	Exact parent algebra with fail-closed radial crosswalk	Covariant parent and transverse-traceless sector; equality with the intrinsic radial τ-jet requires a projective-cocycle comparison
Euler cut current and Einstein packet isotropy	Exact symbolic and reduced-mode theorem with mutation tests	Finite-radius smooth compact-frequency core; endpoint interchange conditional
Ricci–Bach composition and exact sequence	Exact symbolic theorem	Schwarzschild, linearized, declared mode classes
RW/RW/spin-one filtration and partial jet	Exact symbolic theorem with independent certificate	Axial $\ell = 2$
No rational spin-one/spin-two intertwiner	Exact symbolic theorem with mutation tests	Axial $\ell = 2$, fixed real $\omega > 0$
RW/Maxwell simplicity, scalar endomorphisms, and axial local-C obstruction	Exact symbolic theorem with independent certificate	Every integer $\ell \geq 2$ and real $\omega > 0$ for the scalar factors; the nonsplit Bach involution conclusion is axial $\ell = 2$
Complete dual-number commutant of the repeated-spin-two block	Exact categorical and symbolic theorem with independent certificate	Axial $\ell = 2$, every real $\omega > 0$; not the full six-state commutant
Positive-graph obstruction and $B_2/E \simeq E^*$	Exact endpoint Krein theorem	Axial spin-two endpoint trace space; pointwise packet graph form only for decomposable frequency-invariant subspaces
Spectral fundamental symmetry and weighted threshold completion	Exact finite-dimensional functional-calculus theorem	Each real-frequency incoming fiber and compact positive-frequency bands; no canonical, causal, holomorphic, or BRST claim
Incoming factor Gram and $T_- \in GL(3,\mathbb{C})$	Exact plus computer-assisted theorem	Every real $\omega > 0$, $M = 1$ normalization
Outgoing population and pseudo-isometry	Validated interval theorem	I_*; genericity off a locally finite set
Exact all-ℓ scalar threshold non-resonance	Exact symbolic theorem	Spin one and spin two, every integer $\ell \geq 2$; no punctured zero-free scattering interval claimed
No growing separated mode	Analytic filtered energy theorem	Axial $\ell = 2$, declared complex boundary classes
Connection Smith valuations $(0, 0, 2)$	Validated projective Evans and local-unit theorem	One enclosed simple damped QNM
QNM divisor count $N_B = 2N_2 + N_1$	Exact triangular determinant theorem	Contours avoiding scalar zeros, poles, and endpoint-normalization singularities
Second-order Green-resolvent pole	Open conditional corollary	Requires analytic Fredholm realization
Time-domain boundedness and decay	Open	Requires threshold, limiting-absorption and resolvent bounds

13 Conclusions

The Einstein sector of pure Weyl gravity is a null bulk sector, not a positive subtheory with the inherited Hessian. Making an Einstein direction nondegenerate requires a partner and thereby produces an indefinite hyperbolic plane. In the Schwarzschild problem the additional carrier is not removed by the ordinary one-ended Lorentzian endpoint conditions tested here. The result is stronger than a local degree-of-freedom count:

- the pure-Weyl bulk Hessian factors universally through linearized Ricci curvature on Ricci-flat backgrounds;
- the trace-adjusted carrier is a constrained linearized Einstein field, whose target-gauge sector is a wrong-sign Maxwell field;
- a second-order auxiliary-tensor parent action makes the source/target Einstein pairing and its canonical null lift explicit;
- its exact inverse expresses the metric response as two Einstein resolvents separated by one algebraic trace-reversal insertion;
- the Einstein packet self-current is an Euler cut contribution, so the Einstein wave-packet subspace is isotropic but nonradical;
- no uniformly positive closed endpoint subspace contains the pure Einstein channel, and the spin-two quotient is canonically dual to the Einstein line;
- the carrier belongs to an exact non-split filtered module;
- neither the spin-one quotient nor the repeated spin-two partner can be reorganized into a positive local branch decomposition by the rational maps tested here;
- the complete local commutant of the repeated-spin-two block is the dual-number algebra, so it has no nontrivial local semisimple branch observable;
- a nonlocal spectral fundamental symmetry nevertheless exists on every positive-real-frequency solution fiber and on every compact band, with a weighted rather than unweighted threshold completion;
- it enters the current derived from the Weyl action;
- the complete incoming flux space is nondegenerate and indefinite;
- every incoming direction is populated by a future-horizon-regular solution at every positive real frequency;
- outgoing population holds on a certified band and generically on the positive axis;
- the full filtered system has no growing separated mode;
- one physical spin-two QNM is a defective connection-level double root.

Thus negative classical flux directions coexist with separated-mode stability. The remaining threat to time-domain control is quantitative resolvent or pseudospectral growth, not an exponentially growing separated frequency.

The publication theorem above is closed without any of the following strengthenings. The first scattering extension is to compute the explicit typed T_+ and reflection matrix over a substantial real-frequency band using a better-conditioned correlated partial-jet transport and then evaluate the exact C-intertwining defect of Proposition 7.3. A separate spectral extension is to certify a global non-Einstein solution satisfying the simultaneous future-horizon-regular and pure-outgoing quasinormal conditions, including a nonzero carrier reconstruction. Further work should derive the finite-mass axial carrier system and compare $[\mathcal{I}_{\mathrm{mass}}]$ with $[\mathcal{I}_{\mathrm{Bach}}]$; only that test can promote the intrinsic τ-jet to a physical mass derivative. Finally, construct the analytic Fredholm QNM realization needed to promote Theorem 10.2 to a second-order Green-resolvent pole. The parent formulation gives a cheaper audit inside that task: compare $\langle \widetilde{f}_n, \mathcal{A} f_n\rangle$ with the radial extension overlap before attempting a full resolvent promotion. A separate time-domain control should compare direct six-state evolution with the sequential retarded convolution of Remark 3.6. In parallel, the polar problem should be reorganized around the standard Zerilli–Moncrief map and the target-gauge Maxwell scalar, so that only the lift, endpoint units, and extension shears are computed afresh.

A Typed endpoint frames and explicit Grams

The triangular factor frame and the diagonal signature frame serve different purposes. After normalizing the repeated-spin-two hyperbolic plane, write

$$\mathcal{B}_0 = (E, R, S), \qquad J_0 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The corresponding signature frame is

$$P = \frac{E+R}{\sqrt{2}}, \qquad N = \frac{E-R}{\sqrt{2}}, \qquad S = S,$$

with

$$J_\sigma = \operatorname{diag}(1,-1,-1), \qquad C = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad C^\dagger J_0 C = J_\sigma.$$

If M_0 is written in the factor frame, then

$$M_\sigma = C^{-1} M_0 C.$$

For

$$M_0 = \begin{pmatrix} m & x & y \\ 0 & m & z \\ 0 & 0 & n \end{pmatrix},$$

one finds

$$M_\sigma = \begin{pmatrix} m + x/2 & -x/2 & (y+z)/\sqrt{2} \\ x/2 & m - x/2 & (y-z)/\sqrt{2} \\ 0 & 0 & n \end{pmatrix}.$$

Thus triangularity is a statement about the filtration frame, not the signature frame. In particular a nonzero extension entry x produces both upper and lower off-diagonal entries after signature normalization.

In the original incoming order $(XI0, XI1, EI0)$, the normalized action Gram is

$$G_-^{\mathrm{raw}} = \begin{pmatrix} \frac{576}{5\omega} & \frac{96i}{5} & \frac{384\omega}{5} \\ -\frac{96i}{5} & -\frac{144\omega}{5} & -\frac{192i\omega^2}{5} \\ \frac{384\omega}{5} & \frac{192i\omega^2}{5} & 0 \end{pmatrix}. \tag{31}$$

The factor shear and orthogonal quotient lift reduce (31) to (18).

For reference, in the original outgoing order $(XI2, XI3, EI2)$, the normalized future Gram is

$$G_+^{\mathrm{raw}} = \begin{pmatrix} \frac{384(-512\omega^6+640\omega^4-278\omega^2+39)}{5\omega} & \frac{12288i\omega^4-3072\omega^3-9984i\omega^2}{5}+192\omega+384i & \frac{-1536i\omega^2+384\omega+768i}{5} \\ \frac{-12288i\omega^4-3072\omega^3+9984i\omega^2}{5}+192\omega-384i & -\frac{768\omega^3}{5}+48\omega & \frac{96\omega}{5} \\ \frac{1536i\omega^2+384\omega-768i}{5} & \frac{96\omega}{5} & 0 \end{pmatrix}. \tag{32}$$

The full physical Grams are $\pi\alpha_{\mathrm{W}}G_\pm^{\mathrm{raw}}$. On $[1/2, 3/4]$, exact rational/Sturm bounds give

	$\max \lVert G_\pm \rVert_F^2$	$\max \lVert G_\pm^{-1} \rVert_F^2$
G_-	1429056/25	7025/65536
G_+	10389888/25	19825/65536.

These estimates give uniform equivalence of the finite-band coefficient norms without inferring inverse bounds from determinants alone.

A.1 The quotient metric is intrinsic

Let K be the repeated-spin-two plane and M_1 the spin-one quotient. Because $G_-|_K$ is nondegenerate,

$$\pi_1 : K^{\perp_{G_-}} \longrightarrow M_1$$

is an isomorphism. If $s_\perp$ denotes its inverse, then

$$g_1(q_1, q_2) = G_-(s_\perp q_1, s_\perp q_2)$$

is independent of every arbitrary lift. In a nonorthogonal adapted frame it is the Schur complement

$$g_1 = G_{11} - G_{1K}G_{KK}^{-1}G_{K1}.$$

The value $g_1 = -32\pi\alpha_{\mathrm{W}}/(15\omega)$ is therefore a quotient-theoretic statement, not a sign attached to one chosen Witt vector.

B Krein scattering identities

In either typed frame define the Krein adjoint

$$M^\sharp = J^{-1}M^\dagger J.$$

After separate endpoint normalizations, the one-sided conservation law is

$$R^\sharp R + A^\sharp A = I.$$

The horizon defect operator and its Hermitian form are

$$\mathfrak{D}_H = I - R^\sharp R = A^\sharp A, \qquad D_H = J - R^\dagger J R = A^\dagger J A.$$

On the certified cell $A = T_-^{-1}$ is invertible and the horizon Gram has inertia $(1,2,0)$. Hence

$$\mathfrak{D}_H \in GL(3,\mathbb{C}), \qquad 1 \notin \sigma(R^\sharp R), \qquad \operatorname{inertia} D_H = (1,2,0).$$

The defect is nondegenerate but indefinite. It is not a positive absorption operator, and the conservation theorem supplies no inequality of the form "outgoing flux is at most incoming flux" on the complete Weyl trace space.

In canonical normalized frames $\det J = 1$, so

$$\det D_H = |\det A|^2 = |\det T_-|^{-2}.$$

Using

$$\det T_- = A_{\mathrm{in},2}^2 A_{\mathrm{in},1}$$

and the scalar Wronskians,

$$\det D_H = |A_{\mathrm{in},2}|^{-4}|A_{\mathrm{in},1}|^{-2}, \qquad 0 < \det D_H \le 1.$$

This positive signed three-volume is the surviving determinant-level remnant of ordinary scalar absorption; the individual defect eigenvalues need not lie in $[0,1]$.

An algebraic two-sided completion also follows. The range of $\mathscr{S} = (R, A)^T$ is nondegenerate with inertia $(1,2)$ in a target of inertia $(2,4)$. Its orthogonal complement has inertia $(1,2)$. Appending a $J \oplus J$-orthonormal basis of that complement gives an element of $U(2,4)$. This is only an algebraic completion; it does not construct the physical past-horizon channels.

C Projective Evans reduction and the certified disk

For a scalar companion system

$$Y_x = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} Y$$

use the projective coordinate $v = Y_2/Y_1$. It satisfies

$$v_x = A_{21} + (A_{22} - A_{11})v - A_{12}v^2.$$

Let v_H and v_∞ be the horizon-regular and infinity-outgoing lines transported to a common section. Their mismatch

$$\delta = v_\infty - v_H$$

obeys the exact scalar equation

$$\delta_x = \big[A_{22} - A_{11} - A_{12}(v_\infty + v_H)\big]\delta.$$

It follows that projective Evans functions at two matching sections differ by an analytic nonzero factor. Their zero divisors, logarithmic contour moments, and local velocity ratios are match-section independent.

If τ denotes the intrinsic partial-jet deformation, the tangent variables satisfy

$$(v_p)_x = F_v v_p + F_p, \qquad p \in \{\omega, \tau\},$$

where F is the Riccati right-hand side. At a simple zero,

$$\kappa_n = \frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n)}{\delta_\omega(\omega_n)}.$$

This removes every homogeneous amplitude from the Smith selector.

On a boundary contour avoiding zeros,

$$\frac{1}{2\pi i} \oint \frac{\delta_\omega}{\delta} \, \mathrm{d}\omega$$

counts scalar QNMs, while

$$\frac{1}{2\pi i} \oint \frac{\delta_\tau}{\delta} \, \mathrm{d}\omega$$

is the sum of their intrinsic velocity classes. A projective chart change is a Möbius transformation. It multiplies δ by an analytic nonzero unit whenever both endpoint lines remain in the chart, so the closed-contour invariants are unchanged.

The certified disk in Theorem 10.2 was closed in two stages. First, validated projective panels excluded zero on the boundary and supplied a lifted argument change of 2π. Second, a local interval-Newton rail excluded zero from δ_ω and enclosed the unique simple root. The intrinsic tangent and the independent spin-one mismatch were then evaluated on the same disk. This separation prevents a numerical root approximation from being reused as its own simplicity or local-unit proof.

D Certificate authorities

The principal machine authorities are:

- `black_hole_programme/certificates/BH2A_AXIAL_OPERATOR.json`;
- `black_hole_programme/phase3/axial_rw_lx_triangular_preflight/certificate.json`;
- `black_hole_programme/phase3/axial_null_flux_gram/certificate.json`;
- `black_hole_programme/phase3/axial_incoming_extended_domain_audit/certificate.json`;
- `black_hole_programme/phase3/axial_boundary_devissage_no_growth/certificate.json`;
- `black_hole_programme/phase3/axial_outgoing_population_cell_half_v1/certificate.json`;
- `black_hole_programme/phase3/axial_global_finite_flux_channel_classification_v3/certificate.json`;
- `black_hole_programme/phase4/axial_threshold_exact_structure_v1/certificate.json`;

- `black_hole_programme/phase4/axial_all_ell_threshold_structure_v1/certificate.json`;
- `black_hole_programme/phase4/axial_universal_hessian_intertwiner_v1/certificate.json`;
- `black_hole_programme/phase4/covariant_einstein_maxwell_carrier_v1/certificate.json`;
- `black_hole_programme/phase4/weyl_euler_current_transgression_v1/certificate.json`;
- `black_hole_programme/phase4/second_order_parent_flux_v1/certificate.json`;
- `black_hole_programme/phase4/parent_resolvent_krein_obstructions_v1/certificate.json`;
- `black_hole_programme/phase4/rw_maxwell_simplicity_endomorphisms_v1/certificate.json`;
- `black_hole_programme/phase4/einstein_weyl_critical_mass_jet_v1/certificate.json`;
- `black_hole_programme/phase4/axial_local_commutant_spectral_c_v1/certificate.json`;
- `black_hole_programme/phase4/axial_local_nonlocal_positivity_v1/certificate.json`;
- `black_hole_programme/certificates/BH1A_NORMALIZED_GENERATOR.json`;
- `black_hole_programme/phase3/axial_qnm_projective_evans_contour_completion/full_contour_winding_v1/certificate.json`;
- `black_hole_programme/phase3/axial_qnm_projective_evans_contour_completion/local_selector_v1/certificate.json`;
- `black_hole_programme/phase3/axial_qnm_spin_one_local_unit_v1/certificate.json`;
- `black_hole_programme/certificates/BH2B_POLAR_REACH.json`.

Each numerical theorem has a separately implemented verifier and mutation tests. Reproduction commands and content hashes are recorded in the generated claim map and receipt accompanying this paper.

E What this paper does not establish

It does not establish:

- explicit T_+ entries or reflection amplitudes over a substantial frequency band;

- a non-Einstein separated mode that is simultaneously future-horizon regular and purely outgoing (or no-incoming) at infinity; the real-frequency title theorem concerns the one-sided trace maps of Definition 1.1;
- absence of isolated positive-real reflection zeros;
- a punctured threshold interval of outgoing invertibility or the formal low-frequency Jost and extension-ratio asymptotics in Section 8;
- a complete asymptotically flat metric/tetrad phase space and charge algebra;
- unconditional interchange of the null or horizon limit with the Einstein wave-packet time integral;
- Euler exactness or removability of the mixed Einstein–additional pairing;
- a polar global connection theorem;
- a generic implication from radial nonsplitting to a time-translation Jordan block;
- a complete classification of the complex-frequency reducibility locus, or the absence of nonlocal, spectral, scattering-dependent, or BRST C operators;
- an analytic Fredholm realization of the damped QNM boundary problem;
- a physical second-order Green-resolvent pole or generalized ringdown term;
- equality of the parent QNM overlap with the normalized radial overlap before scalar-projection, commutator, and endpoint terms are audited;
- a Schwarzschild retarded-convolution evolution theorem or the flat-space biwave control in the curved implementation;
- limiting absorption, uniform resolvent bounds, threshold control, bounded time evolution, decay, or completeness;
- a positive quantum metric, particles, a BRST physical Hilbert space, CPT, or unitarity.

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